

Federico Peiretti, Ph.D.

PERSONAL INFORMATION

Federico Peiretti
Ph.D. in Computer Science

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RESEARCH INTERESTS

- **Trustworthy & Responsible AI:** Algorithmic Fairness, Privacy and Data Protection, Privacy-Preserving Data Mining (anonymization, differential privacy).
- **Machine Learning**
- **Data Mining**
- **Clustering and Co-clustering** for sparse, high-dimensional data.
- **Social Network Analysis:** Social Networks, Privacy Measures, Social Media.

EDUCATION

2022–2026	Ph.D. in Computer Science Department of Computer Science, University of Turin – Turin, Italy Thesis: <i>Fair Associative Co-Clustering</i> . Advisors: Prof. Ruggero G. Pensa, Dr. Sara Capecchi, Dr. Livio Bioglio
2016–2022	M.Sc. in Computer Science Department of Computer Science, University of Turin – Turin, Italy Grade: 110/110 Thesis: <i>Sensitivity Analysis of Italian Social Media Posts with Transformer-Based Language Models</i> . Advisor: Prof. Ruggero G. Pensa

2012–2016 **B.Sc. in Computer Science**
Department of Computer Science, University of Turin – Turin, Italy
Grade: 98/110
Thesis: *Social4School: An Educational Support to Encourage Privacy-Aware Behaviour in Social Networks*.
Advisor: Prof. Ruggero G. Pensa

RESEARCH EXPERIENCE

- 11/2025–04/2026 **Research Fellow**
Computer Science Department, University of Turin - Turin, Italy
Project: DESPITE FRAUD (P.R. FESR 2021/27 SWIch Call).
Research directors: Prof. Claudio Schifanella, Prof. Ruggero G. Pensa.

Conducted research on food fraud detection through the analysis of Near-Infrared (NIR) and Raman spectroscopic data and the development of machine learning and deep learning models for the classification and quality assessment of fresh and thawed fish products. In collaboration with Politecnico di Torino, Istituto Zooprofilattico Sperimentale Piemonte Liguria e Valle d'Aosta, Esselunga, Amadori.
- 11/2022–10/2025 **Doctoral Research**
Computer Science Department, University of Turin - Turin, Italy
Research group: Machine Learning and Data Mining
Advisor: Prof. Ruggero G. Pensa

Conducted research in Machine Learning and Responsible AI, focusing on algorithmic fairness in co-clustering of sparse, high-dimensional data. Defined the fair co-clustering problem to address disparate impact in the representation of protected groups, established its theoretical foundations, and developed novel fairness-aware co-clustering algorithms.
- 05/2025–08/2025 **Visiting Ph.D. Student**
Computer Science Department, Johannes Gutenberg University - Mainz, Germany
Research group: Data Mining
Research director: Prof. Stefan Kramer

Conducted research on algorithmic trade-offs within Machine Learning pipelines, focusing on representation learning, unsupervised learning, and algorithmic fairness.
- 10/2022–12/2022 **Research Collaborator**
CINI Consorzio Interuniversitario Nazionale per l'Informatica - Turin, Italy

Project: Methods for teaching Computer Science in schools (INFO-DIDA).
Research director: Dr. Sara Capecchi.

Research activity on methods for teaching Computer Science in schools. Designed and delivered classroom activities in primary schools to introduce fundamental concepts on computer science, computational thinking, and digital awareness through unplugged activities and educational coding games. Developed teaching materials and provided training and support to primary school teachers.

01/2016–06/2016 **Research Fellow**

Computer Science Department, University of Turin - Turin, Italy

Project: Supporting the enhancement of privacy perception and awareness in on-line social networks. Funded by Fondazione CRT.

Research director: Prof. Ruggero G. Pensa.

Defined and implemented metrics to measure users' privacy levels and assess the risks of sharing sensitive data on social networks.

11/2015–10/2016 **Bachelor's Internship**

Computer Science Department, University of Turin - Turin, Italy

Co-conceived, designed, and developed Social4School, an educational social networking platform promoting privacy awareness among primary and secondary school students. Conducted classroom experiments and contributed to the collection and analysis of research data.

PUBLICATIONS

Ph.D. Thesis

1. **Peiretti, F.** *Fair Associative Co-Clustering*. PhD thesis, University of Turin, Italy, 2026. <https://hdl.handle.net/2318/2153570>

Journal Articles

1. BATTAGLIA, E., **Peiretti, F.**, AND PENZA, R. G. Co-clustering: A survey of the main methods, recent trends, and open problems. *ACM Comput. Surv.* 57, 2 (2025), 48:1–48:33. [doi:10.1145/3698875](https://doi.org/10.1145/3698875)
2. BATTAGLIA, E., **Peiretti, F.**, AND PENZA, R. G. Fast parameterless prototype-based co-clustering. *Mach. Learn.* 113, 4 (2024), 2153–2181. [doi:10.1007/s10994-023-06474-y](https://doi.org/10.1007/s10994-023-06474-y)
3. BIOGLIO, L., CAPECCHI, S., **Peiretti, F.**, SAYED, D., TORASSO, A., AND PENZA, R. G. A social network simulation game to raise awareness of privacy among school children. *IEEE Trans. Learn. Technol.* 12, 4 (2019), 456–469. [doi:10.1109/TLT.2018.2881193](https://doi.org/10.1109/TLT.2018.2881193)

Conference Papers

1. **Peiretti, F.**, AND PENZA, R. G. Fair associative co-clustering. In *Machine Learning and Knowledge Discovery in Databases. Research Track - European Conference, ECML PKDD 2025, Porto, Portugal, September 15-19, 2025, Proceedings, Part I* (2025), R. P. Ribeiro, B. Pfahringer, N. Japkowicz, P. Larrañaga, A. M. Jorge, C. Soares, P. H. Abreu, and J. Gama, Eds., vol. 16013 of *Lecture Notes in Computer Science*, Springer, pp. 282–300. [doi:10.1007/978-3-032-05962-8_17](https://doi.org/10.1007/978-3-032-05962-8_17)
2. FAMIANI, A., **Peiretti, F.**, AND PENZA, R. G. Interpretable fair distance learning for categorical data. In *Machine Learning and Principles and Practice of Knowledge Discovery in Databases - International Workshops of ECML PKDD 2024, Vilnius, Lithuania, September 9-13, 2024, Revised Selected Papers, Part I* (2024), M. Cerrato, D. Kalinauskaitė, M. Lukosevicius, M. Pechenizkiy, and K. Sutienė, Eds., vol. 2558 of *Communications in Computer and Information Science*, Springer, pp. 81–97. [doi:10.1007/978-3-032-25308-8_6](https://doi.org/10.1007/978-3-032-25308-8_6)
3. **Peiretti, F.**, AND PENZA, R. G. Detection of privacy-harming social media posts in italian. In *Security and Privacy in Social Networks and Big Data - 9th International Symposium, SocialSec 2023, Canterbury, UK, August 14-16, 2023, Proceedings* (2023), B. Arief, A. Monreale, M. Sirivianos, and S. Li, Eds., vol. 14097 of *Lecture Notes in Computer Science*, Springer, pp. 203–223. [doi:10.1007/978-981-99-5177-2_12](https://doi.org/10.1007/978-981-99-5177-2_12)

RESEARCH PROJECTS

Participation

2025–2026

DESPITE FRAUD

University of Turin

Development of a portable decision-support platform combining non-destructive NIR and Raman spectroscopy with machine learning for food authenticity and fraud detection, including the identification of fresh and thawed fish products and meat products containing mechanically separated meat. Funded under the P.R. FESR 2021/27 SWIch Call (Support of the Whole Innovation Chain).

Partners: Akronos Technologies s.r.l., Asseltech S.r.l., Informatica System S.r.l., Istituto Zooprofilattico Sperimentale del Piemonte, Liguria e Valle d'Aosta, Dipartimento di Scienza Applicata e Tecnologia (DISAT) - Politecnico di Torino, Dipartimento di Informatica - Università degli Studi di Torino, Esselunga S.p.a., GESCO s.c.a.

2024–2026

TOPML – Trading off Non-Functional Properties of Machine Learning

Johannes Gutenberg University, Mainz, Germany

Research on trade-offs among non-functional properties of Machine Learning, with a focus on algorithmic fairness, transparency, and privacy in representation learning and unsupervised learning. Funded by the Carl Zeiss Foundation.

2015–2023

Social4School

University of Turin

Research project investigating the use of an interactive social network simulation to promote awareness of online social network dynamics among primary and secondary school students. The platform enables students to experience information-sharing dynamics in a controlled environment and explore issues related to privacy, discrimination, hate speech, and fake news. Funded by Fondazione CRT.

CONFERENCE PRESENTATIONS

2025

ECML PKDD 2025

Paper: "Fair Associative Co-clustering".
Porto, Portugal, September 15-19, 2025.

2025

SEBD 2025

Paper: "An Associative Approach to Fair Co-clustering".
Ischia, Italy, June 16-19, 2025.

2024

ECML PKDD 2024

Poster: "Towards Fair Co-Clustering".
Vilnius, Lithuania, September 9-13, 2024.

2024

EWAF 2024

Poster: "Towards Fair Co-Clustering".
Mainz, Germany, July 1-3, 2024.

2023

NSS-SocialSec 2023

Paper: "Detection of Privacy-Harming Social Media Posts in Italian".
Canterbury, United Kingdom, August 14-16, 2023.

TEACHING ACTIVITIES

University Teaching

Contract Lecturer

2025–2026

Database Analyst and Designer (4 hours)

Law Department, University of Turin - Turin, Italy
I Level Master's Program "Diritto dell'Intelligente Amministrare Pubblico"

2025–2026

Ethics, privacy and AI (4 hours)

Law Department, University of Turin - Turin, Italy
I Level Master's Program "Diritto dell'Intelligente Amministrare Pubblico"

2025–2026

AI Fairness - Data Privacy (8 hours)

Politecnico di Torino - Turin, Italy
II Level Master's Program "HumanAIze"

2023–2024 **AI Fairness - Data Privacy** (4 hours)
Politecnico di Torino - Turin, Italy
II Level Master's Program "HumanAIze"

Academic Tutor

2024–2025 **Database** (20 hours)
Computer Science Department, University of Turin - Turin, Italy
B.Sc. in Computer Science

2022–2023 **Database** (20 hours)
Computer Science Department, University of Turin - Turin, Italy
B.Sc. in Computer Science

Professional Training

2023 **C++ Software Development**
Adgenera Srl, Corporate Academy Bosch TEC

2022 **Digital Literacy**
Gi Group SpA, Asei Academy Srl - Turin, Italy

Professional training on digital literacy, equipping adult learners with essential ICT skills for the workplace and everyday use of digital technologies.

2019 **Digital Literacy**
Adecco Italia SpA, Formazione Lavoro Italia Srl - Turin, Italy

Professional training on digital literacy, equipping adult learners with essential ICT skills for the workplace and everyday use of digital technologies.

2018 **Spreadsheets**
Adecco Italia SpA, Gruppo Sapio Srl - Collegno, Italy

STUDENT SUPERVISION

M.Sc. and B.Sc. Thesis Co-Supervision

2024–2025 [Teaching Artificial Intelligence in High School, from guidelines to the experimentation of a learning path to look inside the black box.](#)
Giuseppina Morena.

2024–2025 [Detection of Privacy-Violating Images: A Classification-Based Approach.](#)
Luca Traversa.

2024–2025 [Detection of Privacy-Violating Images: An Approach Based on Sensitive Object Recognition.](#)
Matteo Tomaselli.

2023–2024	Image Annotation and Machine Learning Models for Potentially Privacy-Sensitive Object Classification. Luca Schiavon.
2023–2024	Image Annotation and Deep Learning Models for Privacy-Sensitive Object Recognition. Roberto Francesco Pegoraro
2023–2024	Image Annotation and Recognition of Privacy-Sensitive Objects. Martina Audisio.
2022–2023	Design of Learning Activities for Introducing Machine Learning to High School Students. Federica Longo, Antonella Sicoli.

SCIENTIFIC ACTIVITIES

Academic Governance

2024–2026	Member of the Ph.D. Programme Monitoring and Review Committee Computer Science Department, University of Turin
2023–2026	Ph.D. Student Representative (38th Cycle) Computer Science Department, University of Turin. Department Council and Ph.D. Board.

Reviewer

Journal	International Journal of Educational Research Open (IJEDRO)
Conference	Algorithmic Fairness through the lens of Metrics and Evaluation (AFME) workshop

CONFERENCE VOLUNTEERING

2024	EWAF 2024 European Workshop on Algorithmic Fairness – Mainz, Germany.
2023	ECML PKDD 2023 European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases – Turin, Italy.

AWARDS & SCHOLARSHIPS

2020–2021	Cisco Cybersecurity Scholarship Cisco Networking Academy
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SUMMER SCHOOLS & TRAINING

2023 **BigDat & DeepLearn 2023 Summer**
7th International School on Big Data & 10th International Gran Canaria School
on Deep Learning.
Las Palmas de Gran Canaria, Spain.

TECHNICAL SKILLS

Programming: Python, Java, C, Kotlin.

Machine Learning and Deep Learning: scikit-learn, PyTorch, TensorFlow, Keras, NumPy, Pandas, NLTK, spaCy.

Trustworthy & Responsible AI: Fairlearn, XACML, ARX Data Anonymization Tool.

Social Network Analysis: NetworkX, Gephi.

Databases: SQL, MySQL, PostgreSQL, MongoDB, GraphQL.

Web Development: PHP, HTML, XML, JSON, JavaScript, jQuery, AJAX, CSS, REST APIs.

Amazon Web Services: AppSync, S3, DynamoDB, Amplify, Cognito.

Tools & Platforms: Linux, Windows, Git, Jupyter, L^AT_EX.

LANGUAGES

Italian Native

English B2

Spanish B1

August 29, 2026